

▼ **Figure 31.23 Fungus-gardening insects.** These leaf-cutting ants depend on fungi to convert plant material to a form the insects can digest. The fungi, in turn, depend on the nutrients from the leaves the ants feed them.



studying how fungal endophytes affect a woody plant tested whether leaf endophytes benefit seedlings of the cacao tree, *Theobroma cacao*. Their findings show that the fungal endophytes of woody flowering plants can play an important role in defending against pathogens.

Fungus-Animal Mutualisms

As mentioned earlier, some fungi share their digestive services with animals, helping break down plant material in the guts of cattle and other grazing mammals. Many species of ants take advantage of the digestive power of fungi by raising them in “farms.” Leaf-cutter ants, for example, scour tropical forests in search of leaves, which they cannot digest on their own but carry back to their nests and feed to the fungi (**Figure 31.23**). As the fungi grow, their hyphae develop specialized swollen tips that are rich in proteins and carbohydrates. The ants feed primarily on these nutrient-rich tips. Not only do the fungi break down plant leaves into substances the insects can digest, but they also detoxify plant defensive compounds that would otherwise kill or harm the ants. In some tropical forests, the fungi have helped these insects become the major consumers of leaves.

The evolution of such farmer ants and that of their fungal “crops” have been tightly linked for over 50 million years. The fungi have become so dependent on their caretakers that in many cases they can no longer survive without the ants, and vice versa.

Lichens

A **lichen** is a symbiotic association between a photosynthetic microorganism and a fungus in which millions of photosynthetic cells are held in a mass of fungal hyphae. Lichens grow on the surfaces of rocks, rotting logs, trees, and roofs in various forms (**Figure 31.24**). The photosynthetic partners are unicellular or filamentous green algae or cyanobacteria. The fungal component is most often an ascomycete, but some

▼ **Figure 31.24 Variation in lichen growth forms.**



◀ A fruticose (shrublike) lichen



▶ A foliose (leaflike) lichen



◀ Crustose (encrusting) lichens

glomeromycete and basidiomycete lichens are known. Recent studies have found that many lichens also have a basidiomycete yeast as a second fungal component. As the role of these yeasts remains unknown, our discussion will focus on the primary fungal partner.

The fungus usually gives a lichen its overall shape and structure, and tissues formed by hyphae account for most of the lichen’s mass. The cells of the alga or cyanobacterium generally occupy an inner layer below the lichen surface (**Figure 31.25**). The merger of fungus and alga or cyanobacterium is so complete that lichens are given scientific names as though they were single organisms. As might be expected of such “dual organisms,” asexual reproduction as a symbiotic unit is common. This can occur either by fragmentation of the parental lichen or by the formation of **soredia** (singular, *soredium*), small clusters of hyphae with embedded algae (see Figure 31.25). The fungi of many lichens also reproduce sexually.

In most lichens, each partner provides something the other could not obtain on its own. The alga or cyanobacterium provides carbon compounds; a cyanobacterium also fixes nitrogen (see Concept 27.3) and provides organic